

A Counterexample to Necessity of Almost-Homogeneous Difference Sets

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2026-06-23

Status

This is a candidate counterexample packet, likely valid pending expert review. It gives a negative answer to Question 4.3 of Margaris, *Almost bi-Lipschitz embeddings using covers of balls centred at the origin*, arXiv:2001.02607, under the literal reading of “linear bi-Lipschitz embeddings into Euclidean space”.

Source Question

Question 4.3 asks whether the condition that $X - X$ is almost homogeneous at 0 is necessary in order to have linear bi-Lipschitz embeddings into some Euclidean space. The source question appears in the final section, immediately after Questions 4.1 and 4.2.

Question 4.1. In Section 3, we construct a prevalent set of linear almost bi-Lipschitz embeddings from a subset of Banach space X into some Euclidean space, when $X - X$ is almost homogeneous at 0. Is it possible to extend the theorem for doubling subsets of Banach spaces?

Question 4.2. Suppose X is a doubling subset of a Banach space. Can we find an almost bi-Lipschitz embedding ψ from X into another Banach space Y such that the set of differences $\psi(X) - \psi(X)$ into Y is almost homogeneous at 0? This would yield a positive answer to the question above.

Question 4.3. In Section 2, we show that if $X - X$ is almost homogeneous and can be embedded into a Hilbert space, using a linear almost bi-Lipschitz map Φ , then $\Phi(X) - \Phi(X)$ is almost homogeneous at 0. This leads to the following question. Is the condition that $X - X$ is almost homogeneous at 0 necessary in order to have linear bi-Lipschitz embeddings into some Euclidean space? (we already know by Theorem 3.4 that it is sufficient).

Idea of the Proof

The tempting positive argument would be: a finite-dimensional image has controlled covering numbers, so pull those covers back to $X - X$. The flaw is that a linear map may be bi-Lipschitz on X without being bi-Lipschitz on the metric space $X - X$. A pair of difference vectors need not itself be a difference of two points of X .

We exploit exactly this gap. Build X as the graph of a Lipschitz curve over the interval. Projection onto the interval is then a linear bi-Lipschitz embedding into $\mathbb{R}$. But at scale $h_n = 2^{-n^3}$, the curve has 2^n tiny straight pieces pointing in mutually orthogonal directions. The corresponding difference vectors lie near the origin, are mutually separated at scale h_n , and are far too numerous for any almost-homogeneous logarithmic covering bound.

Counterexample

Theorem. *There is a compact subset X of the Hilbert space $\mathbb{R} \oplus_2 \ell_2$ such that the coordinate projection*

$$P(t, u) = t$$

is a linear bi-Lipschitz embedding of X into $\mathbb{R}$, but $X - X$ is not (α, β) -almost (M, s) -homogeneous at 0 for any finite parameters $M \geq 1$, $s \geq 0$, and $\alpha, \beta \geq 0$.

Proof. Let

$$m_n = 2^n, \quad h_n = 2^{-n^3} \quad (n \geq 2).$$

Since $\sum_{n \geq 2} m_n h_n < 1$, choose pairwise disjoint intervals

$$I_{n,k} = [a_{n,k}, a_{n,k} + h_n] \subset [0, 1], \quad 1 \leq k \leq m_n.$$

Let $(e_{n,k})_{n \geq 2, 1 \leq k \leq m_n}$ be an orthonormal family in ℓ_2 . Define a measurable function $g : [0, 1] \rightarrow \ell_2$ by

$$g(t) = e_{n,k} \quad \text{for } t \in I_{n,k}, \quad g(t) = 0 \quad \text{otherwise,}$$

and set

$$f(t) = \int_0^t g(u) du.$$

Then $\|g(t)\| \leq 1$, so f is 1-Lipschitz. Put

$$X = \{(t, f(t)) : 0 \leq t \leq 1\} \subset \mathbb{R} \oplus_2 \ell_2.$$

This is compact because it is the graph of a continuous map on a compact interval.

The coordinate projection $P(t, u) = t$ is bounded and linear. For $x_t = (t, f(t))$ and $x_u = (u, f(u))$ in X ,

$$|t - u| \leq \|x_t - x_u\| = (|t - u|^2 + \|f(t) - f(u)\|^2)^{1/2} \leq \sqrt{2} |t - u|,$$

because f is 1-Lipschitz. Hence $P|_X : X \rightarrow \mathbb{R}$ is bi-Lipschitz.

It remains to show that the difference set is not almost homogeneous at the origin. For each $n \geq 2$ and $1 \leq k \leq m_n$, take the two endpoints of the interval $I_{n,k}$. Their difference in X is

$$z_{n,k} = (h_n, f(a_{n,k} + h_n) - f(a_{n,k})) = (h_n, h_n e_{n,k}) \in X - X.$$

Thus $\|z_{n,k}\| = \sqrt{2} h_n$. For $k \neq j$,

$$\|z_{n,k} - z_{n,j}\| = h_n \|e_{n,k} - e_{n,j}\| = \sqrt{2} h_n.$$

Set $r_n = 2h_n$ and $\rho_n = h_n/4$. Then all $m_n = 2^n$ points $\{z_{n,k}\}_{k=1}^{m_n}$ lie in $(X - X) \cap B(0, r_n)$, and no ball of radius ρ_n can contain two of them. Consequently

$$N((X - X) \cap B(0, r_n), \rho_n) \geq 2^n.$$

Suppose, toward a contradiction, that $X - X$ were almost homogeneous at 0. Then for some fixed M, s, α, β and all $0 < \rho < r$,

$$N((X - X) \cap B(0, r), \rho) \leq M \left(\frac{r}{\rho}\right)^s \text{slog}(r)^\alpha \text{slog}(\rho)^\beta.$$

Using $r_n/\rho_n = 8$, $r_n = 2^{1-n^3}$, and $\rho_n = 2^{-2-n^3}$, the right-hand side at (r_n, ρ_n) is bounded by

$$M 8^s \text{slog}(r_n)^\alpha \text{slog}(\rho_n)^\beta \leq C n^{3(\alpha+\beta)}$$

for a constant C depending only on the proposed parameters. This is eventually smaller than 2^n , contradicting the lower bound above. Hence $X - X$ is not almost homogeneous at 0 for any finite parameters. $\square$

Verification Notes

The construction is a compact Lipschitz graph, so no completion or closure issue is hidden. The projection is bi-Lipschitz on X with distortion at most $\sqrt{2}$. The obstruction to almost homogeneity uses a fixed covering ratio $r_n/\rho_n = 8$, so logarithmic corrections can contribute only powers of n , while the number of separated difference vectors is 2^n .

The packet answers only Question 4.3 as worded. It does not settle Question 4.1 about all doubling subsets of Banach spaces, nor Question 4.2 about finding a nonlinear almost bi-Lipschitz preprocessing map whose image difference set is almost homogeneous at 0.

Novelty Check

The cheap run indexes were searched for arXiv:2001.02607, the title, Margaris, Olson–Robinson, **almost homogeneous at 0**, and **linear bi-Lipschitz**. Local source search and a targeted web search for the exact Question 4.3 phrase and close variants found the source paper but no later answer. This is a bounded novelty check, not a comprehensive literature survey.

Human Review Recommendation

Review the interpretation of “linear bi-Lipschitz embedding” in Question 4.3. If it means a bounded linear operator on the ambient Banach space whose restriction to X is bi-Lipschitz, the counterexample gives a negative answer.

References

- [1] A. Margaris, *Almost bi-Lipschitz embeddings using covers of balls centred at the origin*, arXiv:2001.02607.